Technical Report: Smart-Hysteresis Thermal Monitor

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1 Introduction

This report documents the development of a temperature control system based on the **Arduino Uno** platform. The system is designed to provide real-time thermal monitoring with an integrated safety alarm governed by hysteresis logic to ensure signal stability.

2 System Architecture

The project was prototyped using the *Autodesk Tinkercad* environment. The hardware components, all included in the **Elegoo Super Starter Kit**, are:

- **Microcontroller:** Arduino Uno R3.
- **Sensing:** TMP36 Precision Analog Temperature Sensor.
- **Display:** I2C 16x2 Liquid Crystal Display (LCD).
- **Indicators:** Red LED and Passive Piezoelectric Buzzer.

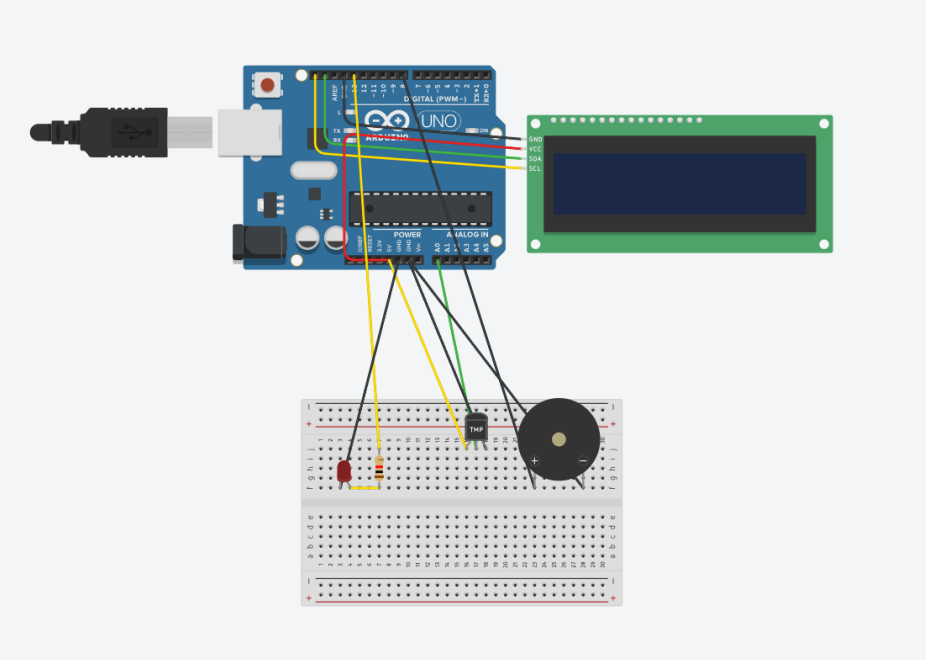


Figure 1: TinkerCad circuit view.

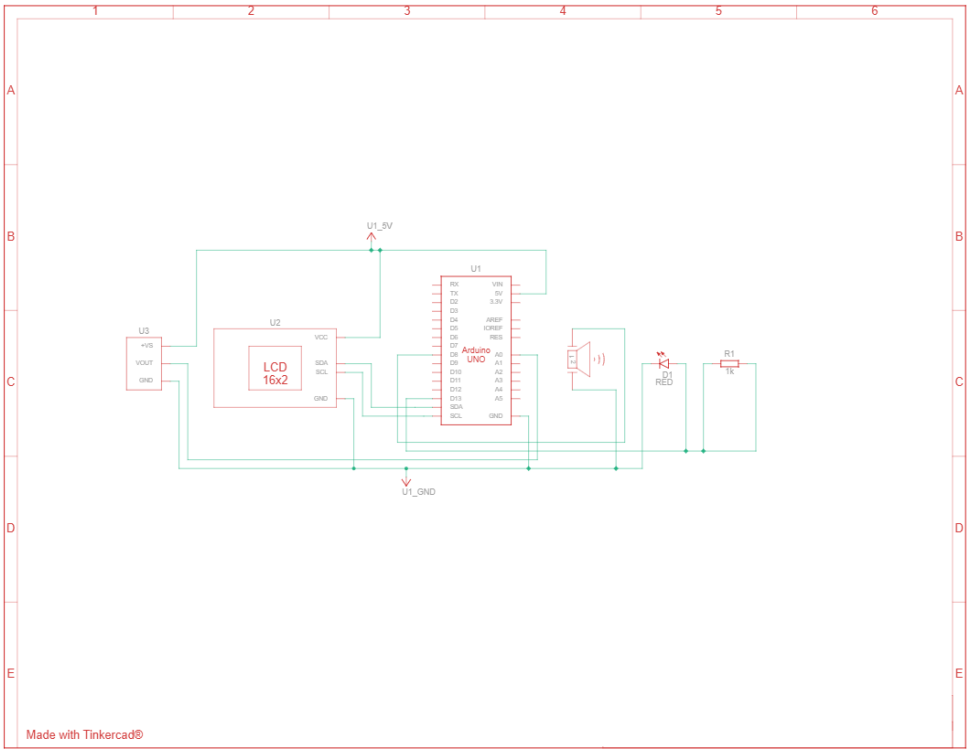


Figure 2: TinkerCad wiring schematic view.

3 Mathematical Modeling

3.1 Analog-to-Digital Conversion

The Arduino ADC converts the analog voltage from the TMP36 sensor into a 10-bit digital value $(0 - 1023)$. The conversion to Celsius degrees follows the transfer function:

$$T(^{\circ}C) = \left(\frac{ADC_{raw} \cdot 5.0}{1024} - 0.5 \right) \cdot 100 \tag{1}$$

This formula accounts for the $500mV$ offset at $0^{\circ}C$ and the $10mV/^{\circ}C$ scale factor typical of the TMP36 sensor.

3.2 Hysteresis Logic

To prevent "chattering" (rapid switching of the alarm due to noise), a **Schmitt Trigger** approach was implemented with two distinct thresholds:

- **Activation (S_{on}):** $30.0^{\circ}C$
- **Deactivation (S_{off}):** $28.0^{\circ}C$

4 Implementation

The control firmware was written in C++. Below is the implementation of the hysteresis logic:

```
// --- HYSTERESIS LOGIC ---
if (temperatureC >= activation) {
    alarmOn = true;  // Trigger alarm
} else if (temperatureC <= deactivation) {
    alarmOn = false; // Reset alarm
}
```

5 Conclusion

The simulation successfully demonstrated that the hysteresis logic provides a stable control signal, effectively filtering out minor temperature fluctuations. This project serves as a foundational example of industrial sensor interfacing and robust software design. This project has real application in every engineering field.